BAMBOO BEAM ANALYTICS

Experimental Overview

Jeff-Mathieu SAINT-AMOUR

Test Setup: Select all | 3PT | 4PT

Reinforcement Group: All

39 Specimens Tested

8 Experimental Groups

2 Available Test Setups

13 Test Group Combinations

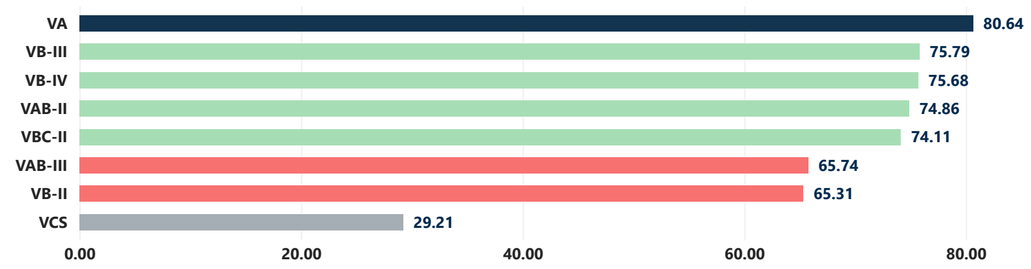
Experimental Coverage

Specimen count by reinforcement group and test setup

Group	3PT	4PT
VA	3	3
VAB-II	3	3
VAB-III	3	3
VBC-II	---	3
VB-II	3	3
VB-III	3	3
VB-IV	---	3
VCS	---	3
Total	15	24

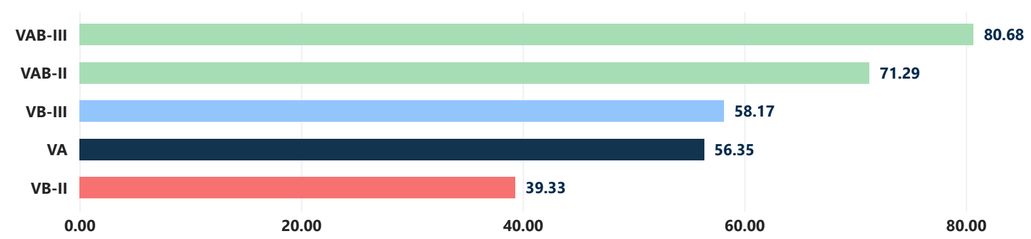
Mean Recorded Load by Group — 4PT

24 specimens • 3 per group



Mean Recorded Load by Group — 3PT

15 specimens • 3 per group



Key Findings

VCS, VB-IV and VBC-II *were tested only in 4PT — **VA:** Steel-only reinforcement reference group — **VCS:** Unreinforced concrete reference group.*

BAMBOO BEAM ANALYTICS

Mechanical Performance

Jeff-Mathieu SAINT-AMOUR

Test Setup: 3PT | 4PT

Reinforcement Group: All

15
Specimens Tested

VAB-III | 80.6...
Highest Group Mean

VA | 2.54%
Lowest Load CV

VB-III | 16.95%
Highest Load CV

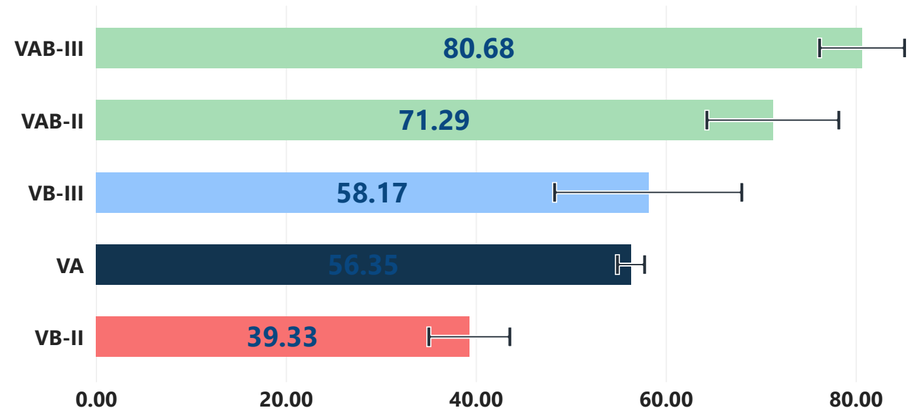
Recorded Load by Reinforcement Group

Comparison with VA — 3PT

Group	Mean Load (kN)	Change vs VA (kN)	Er vs VA (%)
VA	56.35	REF	REF
VAB-II	71.29	+14.94	26.50%
VAB-III	80.68	+24.33	43.17%
VB-II	39.33	-17.02	30.21%
VB-III	58.17	+1.81	3.22%
Total	61.17	+4.81	8.54%

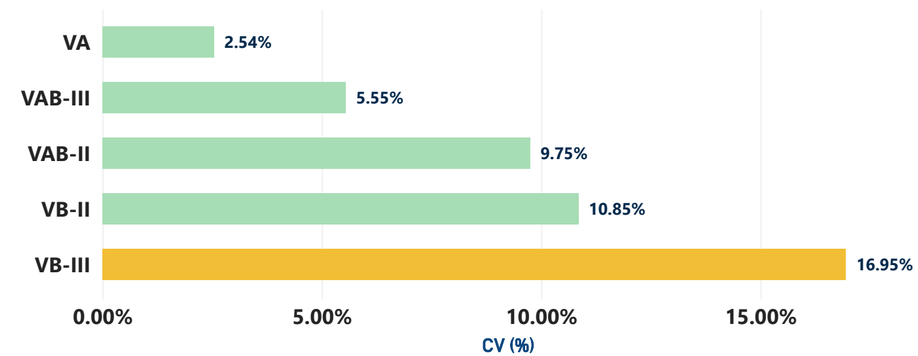
Mean Recorded Load & Variability

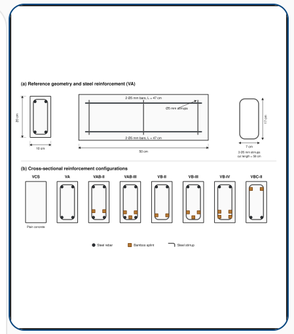
3PT • Mean ± sample SD • n = 3 per group



Relative Load Variability by Group

3PT • Coefficient of variation (CV)





KEY FINDINGS

3PT & 4PT overview
All groups included

01
Highest mean load
3PT: VAB-III — **80.68 kN**
4PT: VA — **80.64 kN**

02
Lowest load CV
3PT: VA — **2.54%**
4PT: VAB-III — **1.72%**

03
Highest load CV
3PT: VB-III — **16.95%**
4PT: VB-IV — **22.07%**

Compare groups within each test setup.

BAMBOO BEAM ANALYTICS

Mechanical Performance

Jeff-Mathieu SAINT-AMOUR

Test Setup: 3PT | 4PT

Reinforcement Group: All

24
Specimens Tested

VA | 80.64 kN
Highest Group Mean

VAB-III | 1.72%
Lowest Load CV

VB-IV | 22.07%
Highest Load CV

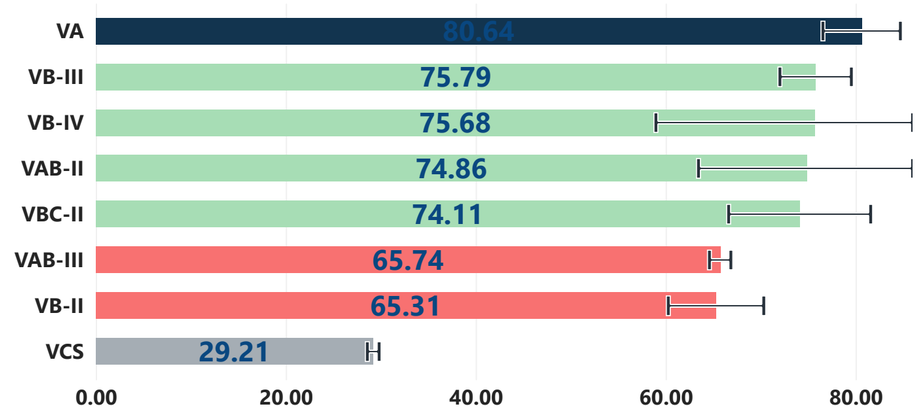
Recorded Load by Reinforcement Group

Comparison with VA — 4PT

Group	Mean Load (kN)	Change vs VA (kN)	Er vs VA (%)
VA	80.64	REF	REF
VAB-II	74.86	-5.78	7.16%
VAB-III	65.74	-14.90	18.48%
VBC-II	74.11	-6.53	8.10%
VB-II	65.31	-15.33	19.01%
VB-III	75.79	-4.85	6.01%
VB-IV	75.68	-4.96	6.16%
VCS	29.21	-51.43	63.77%
Total	67.67	-12.97	16.09%

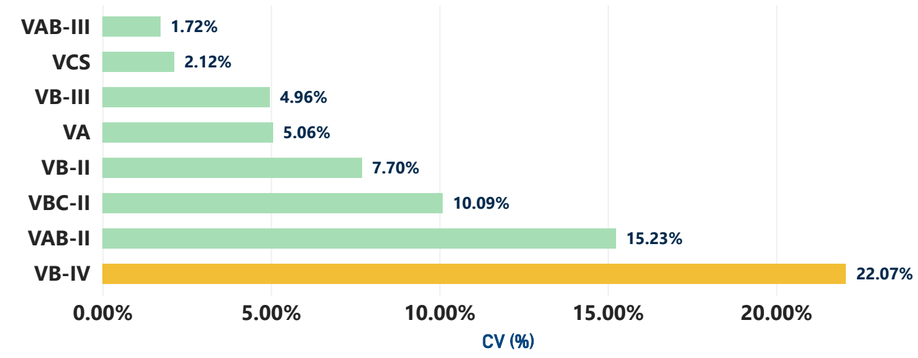
Mean Recorded Load & Variability

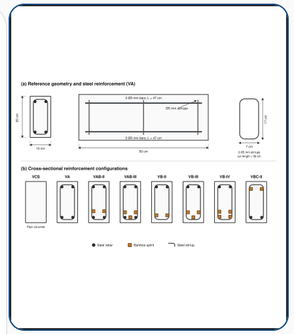
4PT • Mean ± sample SD • n = 3 per group



Relative Load Variability by Group

4PT • Coefficient of variation (CV)





KEY FINDINGS

3PT & 4PT overview
All groups included

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Highest load CV
3PT: VB-III — **16.95%**
4PT: VB-IV — **22.07%**

Compare groups within each test setup.

BAMBOO BEAM ANALYTICS

Cost & Decision Analysis

Jeff-Mathieu SAINT-AMOUR

Test Setup: 3PT | 4PT

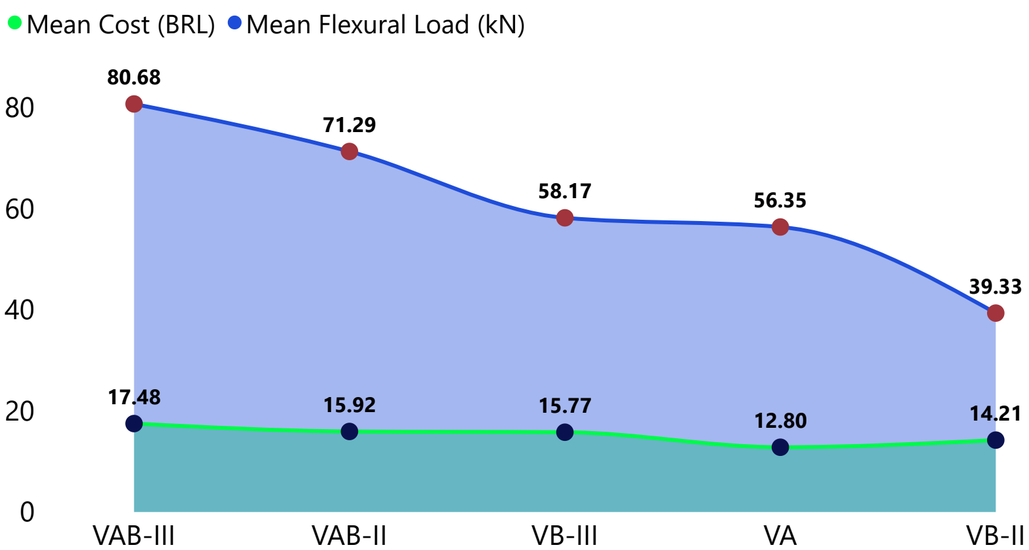
Reinforcement Group: All

VAB-III | 80.68 kN
Top Bamboo Candidate Group

VAB-III | 4.61 kN/BRL
Best Bambo Load/Cost ratio

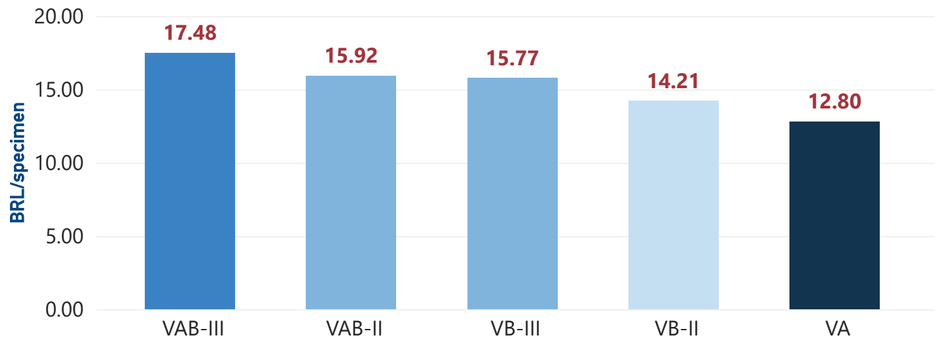
VAB-III | +36.55%
Cost Change vs VA

Mean Recorded Load vs Reinforcement Cost



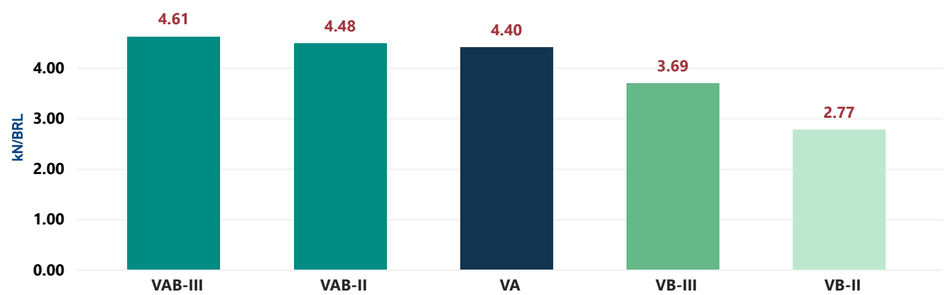
Reinforcement Cost by Group

Mean reinforcement cost (BRL/specimen)



Recorded Load-to-Cost Ratio by Group

Recorded load / reinforcement cost (kN/BRL)



REFERENCE BENCHMARK

- In 3PT, VAB-III's load increase exceeds its cost increase relative to VA, improving the load-to-cost ratio by **4.85%**.
- VA leads the overall load-to-cost comparison at **6.30 kN/BRL** in 4PT.

3PT-KEY FINDINGS

- VAB-III achieves the highest mean recorded load: **80.68 kN**, exceeding VA by **43.17%**.
- VAB-III leads at **4.61 kN/BRL**, compared with **4.40 kN/BRL** for VA.

4PT-KEY FINDINGS

VB-III has the highest bamboo mean load: 75.79 kN.

VBC-II leads bamboo load-to-cost ratio: 5.21 kN/BRL, at 74.11 kN mean load.

BAMBOO BEAM ANALYTICS

Cost & Decision Analysis

Jeff-Mathieu SAINT-AMOUR

Test Setup: 3PT | 4PT

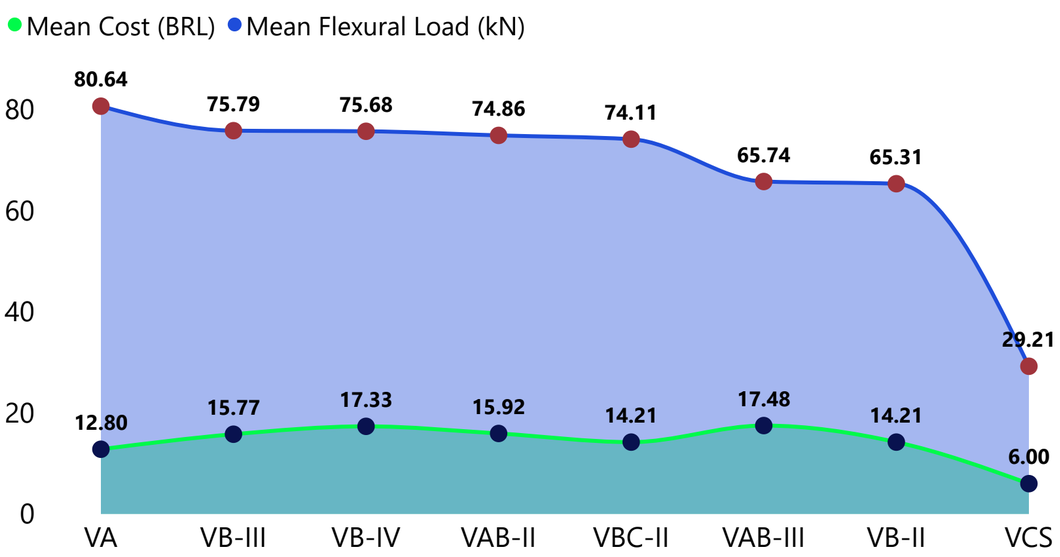
Reinforcement Group: All

VB-III | 75.79 kN
Top Bamboo Candidate Group

VBC-II | 5.21 kN/BRL
Best Bambo Load/Cost ratio

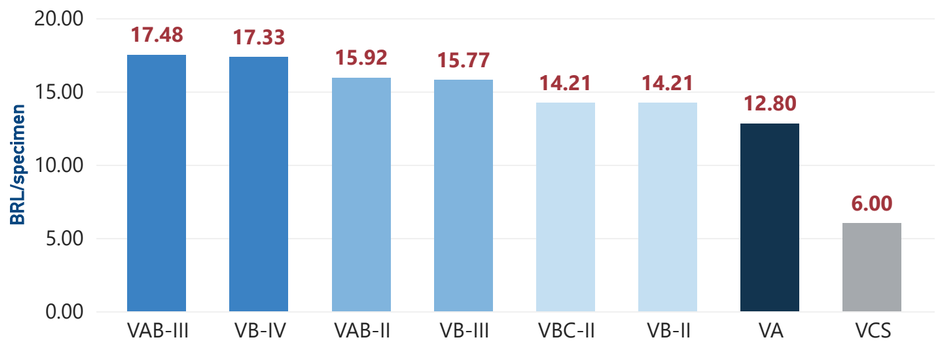
VBC-II | +11.01%
Cost Change vs VA

Mean Recorded Load vs Reinforcement Cost



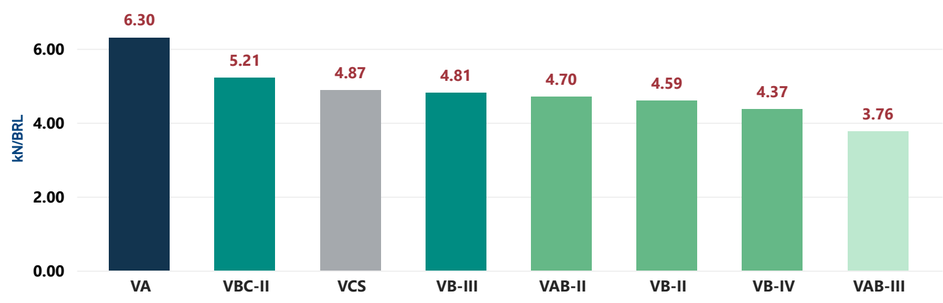
Reinforcement Cost by Group

Mean reinforcement cost (BRL/specimen)



Recorded Load-to-Cost Ratio by Group

Recorded load / reinforcement cost (kN/BRL)



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VB-III has the highest bamboo mean load: 75.79 kN.

VBC-II leads bamboo load-to-cost ratio: 5.21 kN/BRL, at 74.11 kN mean load.

BAMBOO BEAM ANALYTICS

Insights and Conclusions

Jeff-Mathieu SAINT-AMOUR

Test Setup: Select all | 3PT | 4PT

Reinforcement Group: All

39	8	2	13
Specimens Tested	Experimental Groups	Available Test Setups	Test Group Combinations

3PT · MAIN FINDINGS.

01 Highest mean load
VAB-III · 80.68 kN

02 Best bamboo load-to-cost ratio
VAB-III · 4.61 kN/BRL

03 Change vs VA
+43.17% mean load; +36.55% reinforcement cost

VAB-III leads both bamboo indicators.

4PT · MAIN FINDINGS

01 Highest mean load overall
VA · 80.64 kN

02 Highest bamboo mean load
VB-III · 75.79 kN

03 Best bamboo load-to-cost ratio
VBC-II · 5.21 kN/BRL

VBC-II reinforcement cost: +11.01% vs VA

DECISION TAKEAWAYS

- ✓ **3PT:** VAB-III leads bamboo mean load and load-to-cost ratio.
- ✓ **4PT:** VB-III leads bamboo mean load; VBC-II leads bamboo load-to-cost ratio.
- ✓ Consider variability alongside mean load and reinforcement cost.

SCOPE & LIMITATIONS

- **Sample size:** Three specimens per available test–group combination.
- **Comparability:** Compare groups within each test setup. Recorded load is not flexural stress.
- **Scope:** Costs cover reinforcement only. Results do not establish structural design suitability.

VCS, VB-IV and VBC-II *were tested only in 4PT — **VA:** Steel-only reinforcement reference group — **VCS:** Unreinforced concrete reference group.*